Week 6: Balance of Payments

FINA3020: International Finance

Henry Zhang

CUHK Business School

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Week 6 Roadmap: Midterm, then Balance of Payments

Today's two halves

- First 75 minutes: midterm exam on Weeks 1–4, closed book
- 15-minute break
- Second 75 minutes: balance of payments lecture and game

This is the first topic in the second block of the course: national accounts, flows, and institutions.

Appendix overview →

Lecture sections

- **Section 1**: The balance of payments accounts
- **Section 2**: The balance of payments identity
- **Section 3**: Double-entry bookkeeping
- **Section 4**: Global imbalances and the global financial cycle
- **Game 6**: Balance-of-payments ledger relay

What the Balance of Payments Records [Core Concept]

The balance of payments (BoP) records all economic transactions between residents of a country and the rest of the world over a period.

- It is a flow measured over a quarter or a year, not a stock at one date.
- It sorts every transaction into one of three accounts:
  1. **Current account (CA)**: trade in goods and services, primary income, and secondary income.
  2. **Capital account (KA)**: a narrow set of transfers and non-produced, non-financial assets.
  3. **Financial account (FA)**: cross-border purchases and sales of financial assets.
- Every transaction enters twice, once as a credit and once as an equal debit.

We record from an economy's point of view, based on residence.

The Current Account

[Core Concept]

The current account is the sum of four components covering goods, services, primary income, and current transfers between residents and nonresidents.

- **Goods**: Physical trade. The seller is the exporter, the buyer is the importer.
- **Services**: Intangible trade such as travel, transport, finance, and licensing.
- **Primary income**: Interest, dividends, and labor income earned on cross-border factors.
- **Secondary income**: Transfers such as remittances and foreign aid.

US current account by component → | Trade benchmark → | Comparative advantage → | Production / consumption 1 → | Production / consumption 2 → | Deardorff →

The Financial and Capital Accounts

[Core Concept]

The financial account (FA) records transactions in external financial assets and liabilities. Under IMF BPM6, FA balance is **net acquisition of financial assets minus net incurrence of liabilities**.

- A net financial inflow usually means liabilities to nonresidents rise relative to residents' acquisition of foreign assets; in the BPM6 FA balance this is negative (net borrowing).
- A net financial outflow usually means residents acquire foreign assets faster than liabilities rise; in the BPM6 FA balance this is positive (net lending).
- Main types: direct investment, portfolio investment, financial derivatives, other investment (loans, deposits, trade credit), and reserve assets.

The capital account (KA) is narrow under the IMF definition: capital transfers such as debt forgiveness, and transactions in non-produced nonfinancial assets such as transferable contracts, leases, licenses, and marketing assets. KA is usually small.

Terminology: asset trades are often called *capital flows* even though they are in FA, not KA.

Gross vs net flows →

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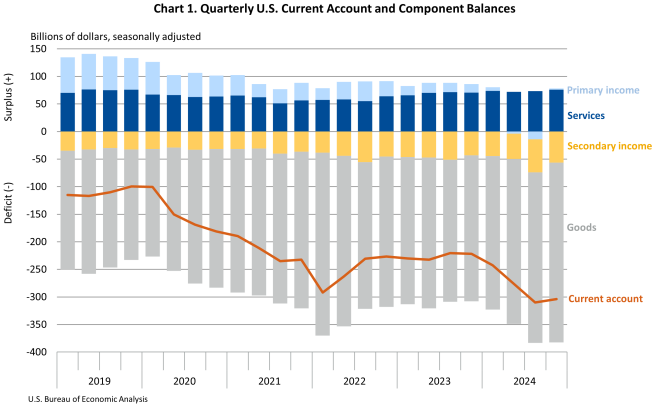
Gross vs net flows →

The US External Accounts

[Background Context]

US current account, 2024	USD tn
Goods balance	−1.22
Services balance	+0.30
Primary income (net)	−0.01
Secondary income (net)	−0.21
Current account balance	**−1.13**

Source: US Bureau of Economic Analysis. Components do not sum exactly to the total because of rounding.



US current account balance from 2019 to 2024

US trade deficit →

The Balance of Payments Identity

[Core Concept]

$$\underbrace{CA}_{\text{current}} + \underbrace{KA}_{\text{capital}} - \underbrace{FA}_{\substack{\text{net acquisition of assets} \\ -\text{net incurrence of liabilities}}} + \underbrace{NEO}_{\text{net errors \& omissions}} = 0$$

This is the IMF BPM6 sign convention. A net financial inflow/net borrowing is a negative FA balance. NEO is the balancing discrepancy in recorded data.

- This is an accounting identity, like GDP = C + I + G + NX. Not causal.
- Ignoring KA and NEO, a current account deficit corresponds to negative FA: net borrowing through greater liability issuance and/or asset sales.
- A current account surplus corresponds to positive FA: net lending through acquisition of foreign assets and/or repayment of liabilities.

Trade and finance → Export financing →

Saving, Investment, and the Current Account

[Core Concept]

Using gross national disposable income, the national accounts give:

$$\text{CA} = S - I$$

where S is national saving and I is domestic investment.

- Abstracting from the capital account, a country that invests more than it saves runs a current-account deficit and is a net borrower from abroad.
- A country that saves more than it invests runs a current-account surplus and is a net lender abroad.
- Split national saving into private and public parts. A larger government deficit lowers S. Holding I fixed, it widens the current account deficit. This is the twin-deficits link.

Derivation from national accounts →

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Derivation from national accounts →

Reserve Assets in the Financial Account

[Core Concept]

Split the BPM6 financial account into nonreserve transactions and reserve-asset transactions:

$$CA + KA - (FA_{other} + \Delta R) + NEO = 0$$

Here ΔR is the net acquisition of reserve assets: reserve accumulation is positive and reserve sales are negative.

- Under a clean float, reserve transactions are usually small.
- Under a peg, if other net inflows do not finance a current-account deficit, the central bank must sell reserve assets to maintain the peg.
- A reserve sale is recorded as negative net acquisition of reserve assets ($\Delta R < 0$). In traditional transaction-entry language, the reduction in a foreign asset is a credit.

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From Flows to Stocks: Net Foreign Assets

[Core Concept]

The balance of payments records flows. These flows accumulate into a stock called the net international investment position, or net foreign assets (NFA):

$$\text{NFA} = \text{foreign assets held by residents} - \text{domestic liabilities to foreigners}$$

- Abstracting from the capital account, valuation effects, and other volume changes, the change in NFA equals the current account.
- The US is a net debtor, yet US net primary income is often close to zero (not very negative). In Weeks 10-11, we will see how the dollar's international role explains this.

Valuation effects and the US position →

Balance of Payments as Double-Entry Bookkeeping [Core Concept]

The balance of payments follows the same double-entry rule as a firm's accounts.

- Every transaction is recorded twice, as a credit and an equal debit.
- One leg is the real or motivating side: the good, service, income, or asset you wanted.
- The other leg is the financing side: how it was paid, whether by cash, a bank deposit, a loan, or a security.
- Total credits equal total debits, so the accounts always balance. The identity is just the sum of all paired entries.

A common error is to record only the visible leg and skip the financing leg.

Credit and Debit Rules

[Core Concept]

These are transaction-entry credits and debits. The published BPM6 financial-account *balance* is then reported as net asset acquisition minus net liability incurrence.

Credit (+): funds come in

- Export of a good or service
- Income or transfer received
- A liability to foreigners rises (you borrow, or a foreigner buys your asset)
- A foreign asset falls (you sell foreign assets or draw down reserves)

Debit (−): funds go out

- Import of a good or service
- Income or transfer paid
- A foreign asset rises (you buy foreign assets or add to reserves)
- A liability to foreigners falls (you repay foreign debt)

Example Ledger: Four Transactions

[Core Concept]

Transaction	Credit	Debit	Amount
Export goods, proceeds held abroad	CA: export	FA: foreign deposit ↑	100
Import goods, funded by a foreign loan	FA: liabilities ↑	CA: import	60
Buy foreign bonds, paid from foreign deposit	FA: foreign deposit ↓	FA: bonds ↑	30
Dividend from abroad, held in foreign deposit	CA: income	FA: foreign deposit ↑	10
Totals	**200**	**200**	

- Current account $= 100 - 60 + 10 = +50$, a surplus.
- BPM6 financial-account balance $= +50$
- CA $-$ FA $= +50 - 50 = 0$

Step-by-step ledger with running positions →

The Second Leg: Do Not Omit the Financing Entry [Core Concept]

- A firm exports 100 of goods, clearly a credit to the current account.
- Consider how the export was paid. The foreign buyer transferred funds, so the exporter now holds a foreign bank deposit.
- That deposit is the second leg: a debit of 100 to the financial account, a financial outflow.
- Every real transaction changes someone's asset or liability. Find that change and record it.

Adding a Reserve Intervention Entry [Core Concept]

Extend the ledger with a central bank that defends a peg.

- Suppose the country runs a current-account deficit of 40, and nonreserve financial transactions are zero.
- To pay foreigners, the central bank sells 40 of foreign reserves to supply the foreign currency.
- In transaction-entry language, reserves $\downarrow 40$ is a financial credit. In BPM6 balance notation, $\Delta R = -40$.
- The identity balances: $\text{CA} - (\text{FA}_{\text{other}} + \Delta R) = -40 - [0 + (-40)] = 0$.

You will learn case studies and crisis dynamics in Week 11.

FDI Earnings in the Balance-of-Payments Ledger [Core Concept]

Use the IMF financial-account sign: acquisition of assets minus incurrence of liabilities. Consider the host economy and a wholly foreign-owned affiliate that earns 100 units of host currency.

Transaction	Current account	Financial account
Retain all profit	Primary income -100	FDI liabilities $+100$, so FA -100
Pay dividend via bank deposits	Primary income -100	Deposit asset falls 100, so FA -100

- Reinvested earnings are imputed income paid to the owner and reinvested as FDI.
- The dividend example assumes payment by reducing the host bank's external deposit asset.
- Neither transaction automatically changes official reserves. Reserve entries require a transaction involving reserve assets controlled by the monetary authority.

Source: IMF, BPM6, chapters 6 and 11.

Greenfield Investment, Acquisitions, and GDP

[Core Concept]

- A foreign parent transfers 1 billion to its host affiliate. The host incurs an FDI liability and obtains a deposit or another financial counterpart.
- The financial transfer alone is not 1 billion of GDP. Construction and equipment purchases enter real investment as production occurs, and imported inputs subtract from net exports.
- A purchase of existing shares from a host resident transfers ownership. The share purchase itself is not newly produced output, although associated services can enter GDP.
- Reminder: residency, ownership, currency, and transaction type are separate.

Source: IMF, BPM6, chapters 6 and 11.

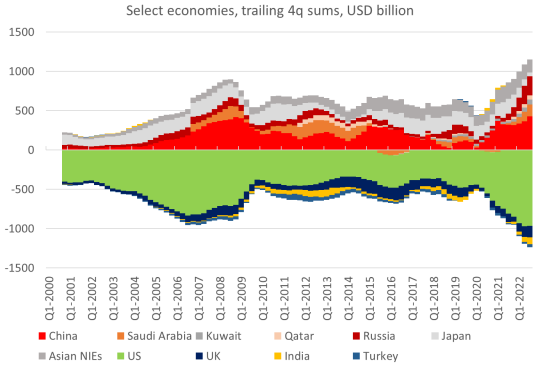
Trade and FDI →

Global Imbalances

[Background Context]

Persistent surpluses in some countries and deficits in others are called global imbalances.

- Surplus side: China, Germany, Japan, and oil exporters run large, persistent surpluses and accumulate foreign assets.
- Deficit side: the US runs persistent deficits, financed by large financial inflows.
- Major Asian economies have run surpluses since the late 1990s. Several built large reserves after the Asian Financial Crisis.
- Imbalances are a concern through risk of sudden reversal of flows (rather than the level)



Current account balances by country group

Source: Setser (CFR); IMF and BEA data

Surpluses in Asia and China → | Gravity model → | Gravity 1 → | Gravity 2 → | Elasticity → | Bilateral → | Cunat–Zymek → | China →

Gross Financial Flows and the Lucas Paradox [Core Concept]

- Gross inflows measure nonresidents' net acquisition of domestic claims. Gross outflows measure residents' net acquisition of foreign claims. Both can rise while net borrowing stays unchanged.
- Each series is itself net of purchases and sales within its category, so a reported gross inflow can become negative during disinvestment.
- The Lucas Paradox asks why capital does not flow more strongly toward capital-scarce economies under a simple diminishing-returns model.
- Human capital, institutions, infrastructure, political risk, and financial frictions affect risk-adjusted returns. A high observed borrowing rate need not indicate a high safe marginal product of capital.

Trade shares → Eaton–Kortum → Trade welfare → Lucas Paradox →

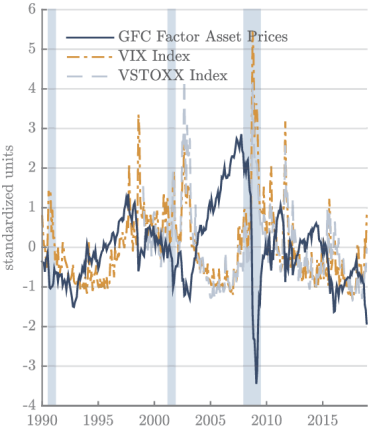
The Global Financial Cycle (GFC)

[Background Context]

Capital flows, asset prices, and credit move together across countries in the GFC.

- Common global factors, often correlated with risk measures such as the VIX, explain an important share of variation in cross-border flows and asset prices.
- US monetary conditions are one important driver given the dollar's central role, alongside global risk appetite, leverage, growth, and commodity shocks.
- Gross flows matter, not only net flows. Inflows and outflows are both large and can reverse together.
- A sudden stop is an abrupt halt or reversal of inflows. It can cause depreciation, credit contraction, and lower output.

Sudden stop mechanism and the Asian crisis → | Financial conditions → | US tightening → | EM financial flows →



GFC factor and the VIX

Source: Miranda-Agrippino & Rey (2022)

Financial Cycles and Trade Cycles

- A common-factor representation is $x_{it} = \lambda_i f_t + \varepsilon_{it}$. The factor summarizes comovement, while loadings describe how each series responds.
- The GFC connects risk appetite, credit conditions, asset prices, and capital flows. It often moves inversely with market volatility measures such as the VIX.
- A global trade and commodity cycle captures common demand and commodity-price movements. It can interact with financial conditions through exporter cash flows and collateral.
- Estimated variance shares and impulse responses depend on the sample and identification. US monetary tightening can reduce external funding, but the magnitude varies by borrower and period.

Regression intuition → OLS diagram → Best linear predictor → Trade / financial cycles →

In-Class Game 6: Balance-of-Payments Ledger Relay [Core Concept]

You oversee the balance of payments ledger for a small open economy.

- **Stage 1**: Scan the QR code. Enter your CUHK ID, name, and section. Read the day's four transactions.
- **Stage 2**: Record both entries for each transaction. Check that total credits equal total debits.
- **Stage 3 [Shock]**: Foreign investors sell domestic equity. Record private dollar supply, the remaining funding gap, and the reserve sale used to defend the peg.
- **Stage 4**: Submit your exit reflection: *"Which second-leg financing entry did your team initially omit, and how did central bank reserve sales balance the ledger under the peg?"*



Open Week 6 Game

Conclusion and Looking Ahead

Today's takeaways

- The balance of payments records cross-border transactions as paired credits and debits.
- Under BPM6, $\text{CA} + \text{KA} - \text{FA} + \text{NEO} = 0$. A deficit corresponds to net financial borrowing.
- $\text{CA} = S - I$ links the external balance to saving and investment.
- Under a peg, reserve sales or purchases may offset the gap between the current/capital accounts and other financial transactions.
- Flows accumulate into net foreign assets, subject to valuation effects.

Looking ahead

- Week 7: international banking and Eurodollar markets.
- The financial account flows we recorded run through bank balance sheets.

Before next class, do the reading and think about how one country's current account deficit shows up as another country's surplus.

Appendix Overview

5 Appendix: Accounts and Data

6 Appendix: Identity and Stocks

7 Appendix: Imbalances & Global Financial Cycle

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Appendix: US Current Account by Component, 2024

[Background Context]

Each component has a gross receipt (credit) and a gross payment (debit). The net figure is the balance.

Component (USD trillion)	Receipts	Payments	Net
Goods	2.08	3.30	−1.22
Services	1.11	0.81	+0.30
Primary income	1.43	1.44	−0.01
Secondary income	0.20	0.41	−0.21
Current account			**−1.13**

- The goods deficit is the largest single item.
- The US sells more services than it buys, driven by finance, travel, and licensing.

Source: US Bureau of Economic Analysis. Figures are rounded, so components do not sum exactly to the total.

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Appendix: Gross vs Net Financial Flows

[Core Concept]

Inflows and outflows are not simply opposite signs of one number. Take the US as an example.

- **Inflow**: the increase in foreigners' holdings of US assets.
- **Outflow**: the increase in US residents' holdings of foreign assets.
- Both are large and positive in a normal year. The financial account records the difference between these flows.
- Only at the global level must gross inflows equal gross outflows, because every asset sold by one country is bought by another.
- All gross flows can both collapse at once: global financial cycle.

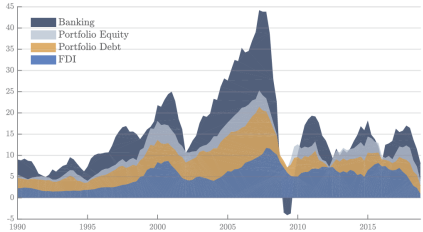


FIGURE 12 International financial flows.

Note: International Flows by asset class as a percentage of world GDP. Annual moving averages. Source IFS. Sample 1990-2018.

Gross cross-border financial flows

Source: Miranda-Agrippino & Rey (2022)

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Appendix: Deriving CA = S − I

[Core Concept]

Start from the national income identity for an open economy:

$$Y = C + I + G + NX$$

- Let net primary income from abroad be NPI and net secondary income be NSI. Gross national disposable income is $Y + \text{NPI} + \text{NSI}$.
- Define national saving as $S = Y + \text{NPI} + \text{NSI} - C - G$.
- Substitute the GDP identity: $S - I = NX + \text{NPI} + \text{NSI} = \text{CA}$.
- Split saving into private and public: $S = \underbrace{(Y + \text{NPI} + \text{NSI} - T - C)}_{\text{private}} + \underbrace{(T - G)}_{\text{public}}$.
- A larger government deficit lowers $T - G$. Holding private saving and investment fixed, the current account deficit widens. This is the twin-deficits link.

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Appendix: Step-by-Step Ledger with Running External Positions [Core Concept]

Track the country's foreign asset and liability stocks as each transaction posts. Start with foreign bank deposits at 0; these are private external assets, not official reserves.

After transaction	Foreign deposits	Bonds	Liabilities
Start	0	0	0
1. Export 100, proceeds to foreign deposit	100	0	0
2. Import 60, foreign loan	100	0	60
3. Buy bonds 30 from foreign deposit	70	30	60
4. Dividend 10 to foreign deposit	80	30	60

- Ending foreign assets $= 80 + 30 = 110$. Ending liabilities $= 60$.
- Net foreign assets rose by $110 - 60 = 50$, exactly the current account surplus.
- The import in row 2 was financed by a foreign loan, so the foreign deposit did not fall. That liability entry is the second leg teams often miss.

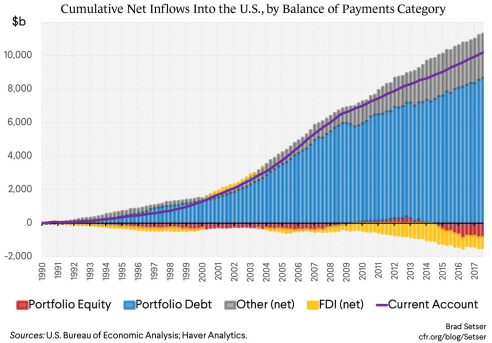
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Appendix: Net Foreign Assets and Valuation Effects

Changes in NFA combine transactions and changes in existing holdings:

$$\Delta \mathrm{NFA} = CA + KA + \text{valuation changes} + \text{other volume changes}.$$

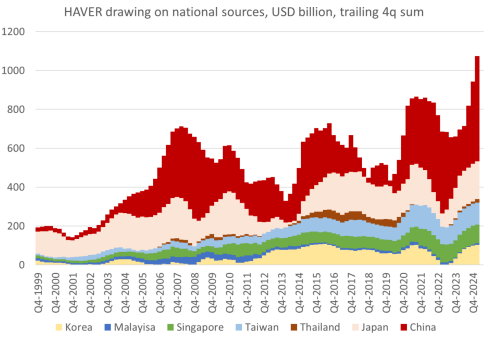
- A rise in foreign asset prices improves NFA, holding liabilities fixed. FX effects depend on the currencies of both assets and liabilities.
- US assets and liabilities differ in composition and returns. Net primary income has sometimes stayed positive despite a negative NFA position.
- Valuation gains are distinct from interest, dividends, and other investment income. The chart records financial-account flows, not the entire change in NFA.



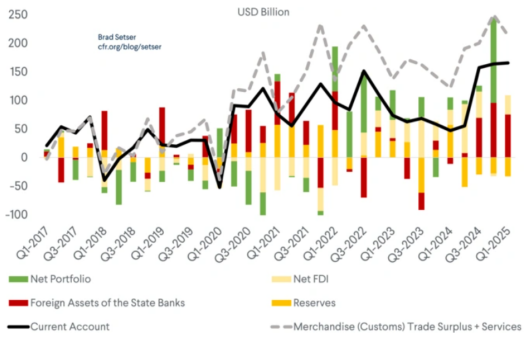
US cumulative financial account

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Appendix: Persistent Surpluses in Asia and China



Current account surpluses across Asia



China's balance of payments

- Several Asian economies have run current account surpluses for over two decades.
- A current-account surplus corresponds to net lending abroad. This may include reserve purchases, private asset purchases, or liability repayment.

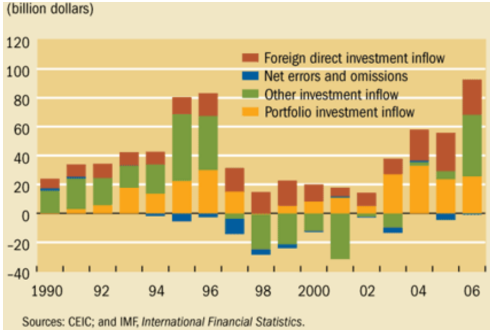
Source: Setser (CFR); data from the IMF

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Appendix: Sudden Stops and the Global Financial Cycle

A sudden stop is an abrupt halt or reversal of financial inflows:

1. A shock lowers global risk appetite. Investors move together toward safe assets.
2. Inflows to emerging markets stop and reverse. The exchange rate depreciates sharply.
3. Borrowers with foreign-currency debt see liabilities rise in local terms. Credit contracts.
4. Output and employment fall. Banks face runs as liquid liabilities flee while assets stay illiquid.



Capital flows in the Asian financial crisis

Source: IMF

Example: net private inflows to five Asian economies swung from about $+93$ billion in 1996 to about -12 billion in 1997. Several pegs broke and the region entered deep recessions.

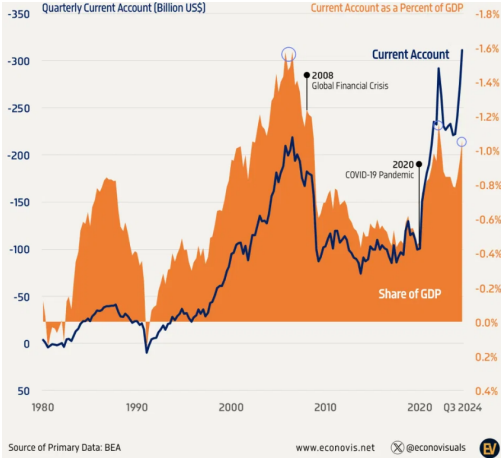
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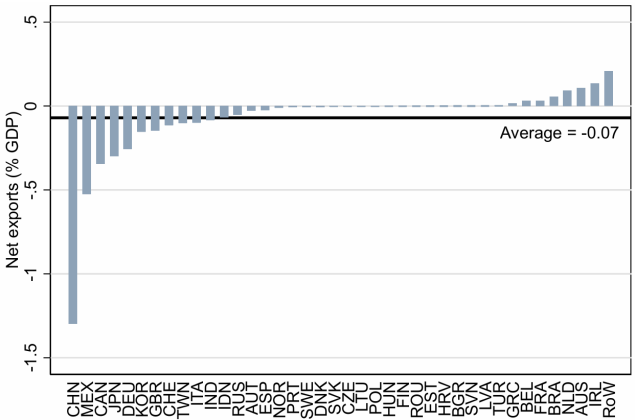
Appendix: Trade Balances and Financial Flows

- An overall current-account balance differs from a bilateral goods deficit because of trade with other partners, services, primary income, and current transfers.
- Comparative advantage and trade costs help explain who trades with whom. Saving and investment decisions help explain aggregate borrowing and lending.
- Trade finance bridges the time between paying workers and receiving export revenue. Dollar invoicing also creates a demand for dollar working capital.
- FDI can substitute for final-goods exports or support cross-border input trade.

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Fact 1: US Trade Deficit

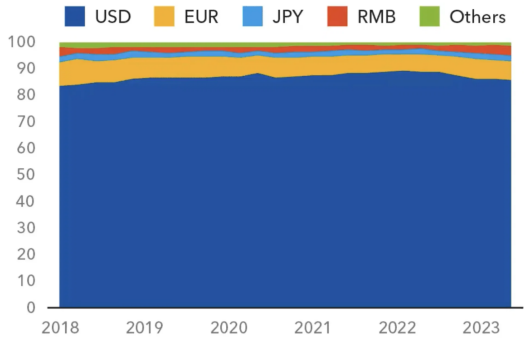




Historical exhibit. Data period, coverage, and units are as labeled.

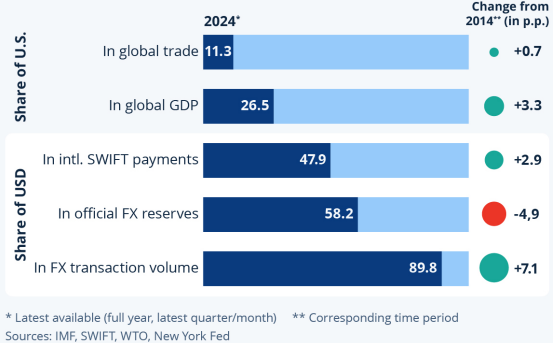
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Fact 2: Trade Flows and Export Financing



Sources: COFER, SWIFT and staff calculations.
Note: Only includes trade finance covered by SWIFT (Code MT700 (Letters of credit)).
Data as of 2023Q3 (FX reserves), 2023Q4 (Trade Finance). Data is exchange-rate adjusted.

IMF



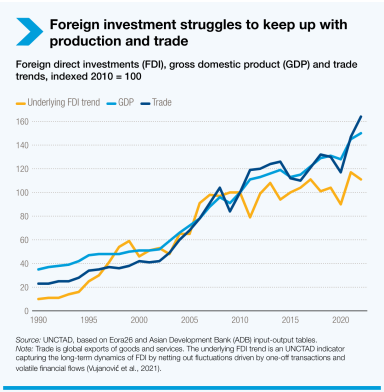
* Latest available (full year, latest quarter/month) ** Corresponding time period
Sources: IMF, SWIFT, WTO, New York Fed

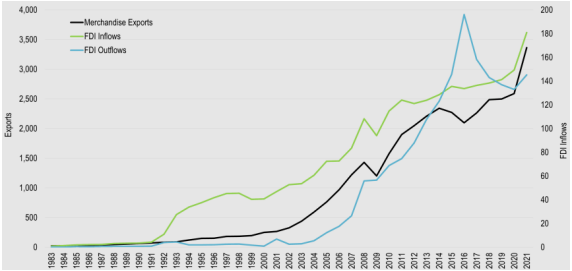
statista

Historical exhibit. Data period, coverage, and units are as labeled.

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Fact 3: Trade Flows and Inbound FDI





Historical exhibit. Data period, coverage, and units are as labeled.

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Appendix: The Benchmark Trade Economy

- Assume competitive goods markets, constant returns, no distortions, and internationally immobile factors for this benchmark.
- Production possibilities describe feasible output. Preferences determine desired consumption. Relative prices coordinate production and consumption choices.
- Goods can cross borders, letting a country consume a bundle different from what it produces. Full autarky excludes both trade and international financial exchange.
- Extensions of the model describe distributional effects, adjustment costs, environmental spillovers, and financing constraints.

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Appendix: Gains from trade and comparative advantage

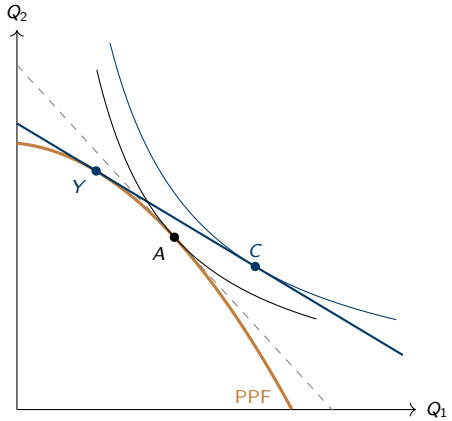
Let p^T be trade prices, $y^A = c^A$ the autarky bundle, and y^T revenue-maximizing production after opening to trade.

$$p^T \cdot y^T \geq p^T \cdot y^A = p^T \cdot c^A.$$

- The representative consumer can afford the autarky bundle at trade prices. Choosing the best affordable bundle therefore gives weakly higher welfare under the benchmark.
- This is an aggregate or representative-agent result. Every individual need not gain without compensation, and strict gains require relevant differences in relative prices or opportunities.
- In the two-good diagram, a lower world price of good 1 than its autarky price encourages home exports of good 2. A higher world price encourages exports of good 1.

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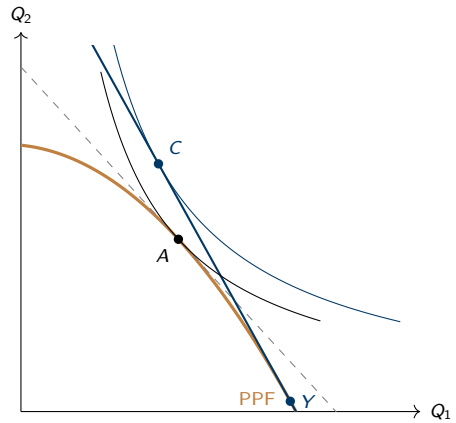
Appendix: Trade Changes Production and Consumption — 1



- In autarky, production and consumption coincide at A. The dashed price line is tangent to the PPF there.
- The blue trade-price line has absolute slope $p_1/p_2 = 0.6$. Production moves to Y, where revenue is highest.
- Consumers choose C on the same budget line and a higher indifference curve. Home exports good 2 and imports the other good.
- Production and consumption differ because goods can be traded.

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Appendix: Trade Changes Production and Consumption — 2



- In autarky, production and consumption coincide at A. The dashed price line is tangent to the PPF there.
- The blue trade-price line has absolute slope $p_1/p_2 = 1.8$. Production moves to Y, where revenue is highest.
- Consumers choose C on the same budget line and a higher indifference curve. Home exports good 1 and imports the other good.
- Production and consumption differ because goods can be traded.

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Appendix: Deardorff's General Comparative Advantage

- Comparative advantage concerns relative opportunity costs. Absolute productivity alone does not determine the pattern of trade.
- The general law relates net exports to autarky prices across goods, subject to the model's assumptions. A country tends to export goods relatively cheap at its autarky prices.
- Observed prices already reflect trade and policy. Autarky prices are counterfactual.

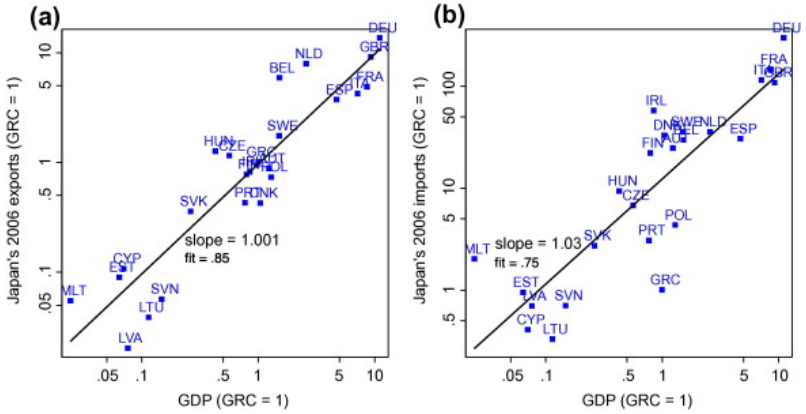
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Appendix: Gravity and Regression Intuition

- Gravity relates bilateral trade to economic size and trade costs. A basic specification is $\log X_{ij} = a + b \log Y_i + c \log Y_j - d \log D_{ij} + u_{ij}$.
- Distance can proxy transport, information, and relationship costs. A correlation between distance and trade need not identify the effect of transportation alone.
- OLS chooses coefficients that minimize squared residuals. It estimates the best linear predictor, which equals the conditional mean only if that mean is linear.
- Exporter and importer effects in structural gravity capture multilateral resistance: bilateral trade depends on alternatives in all markets. Logs exclude zero flows, requiring care in estimation.

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Empirical Fact to Develop Intuition: "Naive" Gravity

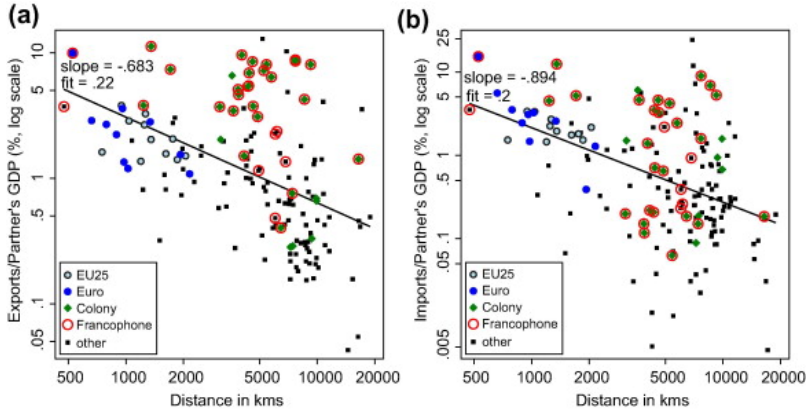


Historical exhibit. Data period, coverage, and units are as labeled.

Source: Head & Mayer (2014).

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Empirical Fact to Develop Intuition: "Naive" Gravity



Historical exhibit. Data period, coverage, and units are as labeled.

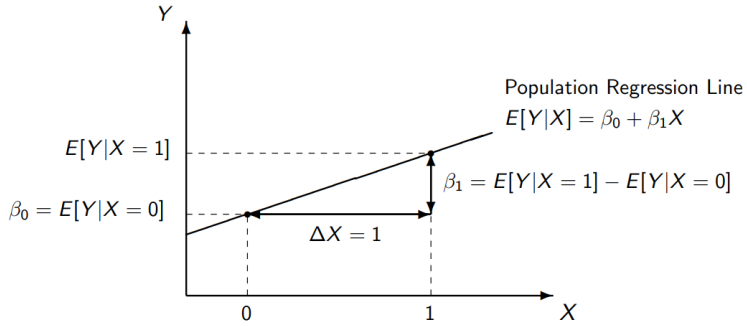
Source: Head & Mayer (2014).

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Digression: Regression and the Conditional Expectation Function

If X is binary, the population regression function is linear:

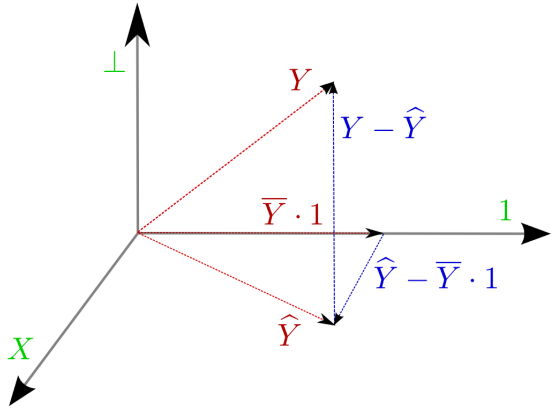
$$E[Y|X] = \beta_0 + \beta_1 X.$$



Historical exhibit. Data period, coverage, and units are as labeled.

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Digression: Visualizing OLS for Linear Regression



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Stanford Stats 191.

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Appendix: Regression as a Best Linear Predictor

Let X_i collect the explanatory variables, including an intercept. The population coefficient is

$$\beta = \arg\min_b E[(y_i - X_i'b)^2] = E[X_i X_i']^{-1} E[X_i y_i].$$

- The inverse requires the *population moment matrix* $E[X_i X_i']$ to be nonsingular. An individual outer product $X_i X_i'$ is generally rank one.
- In a sample, OLS replaces expectations with averages:

$$\widehat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \frac{1}{N} \sum_i X_i y_i.$$

- Sample regressors must have full column rank.
- Distinguish best linear predictor from causality.

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Appendix: From gravity to a distance elasticity

A simple gravity relation gives

$$X_{ni} \approx A Y_n Y_i D_{ni}^{-\gamma}, \qquad \log X_{ni} = \alpha_n + \alpha_i + \zeta \log D_{ni} + \epsilon_{ni}, \quad \zeta = -\gamma.$$

- α_n and α_i are importer and exporter fixed effects, allowing size and other country characteristics to differ.
- If $\widehat{\zeta} = -0.9$, a 1% greater distance is associated with about 0.9% less trade, holding the included country effects fixed.
- More generally, $X_{ni} = A S_i M_n \phi_{ni}$ separates exporter capacity, importer demand, and bilateral accessibility. Armington, Krugman, Eaton–Kortum, and Melitz models can generate gravity relations.
- A current-account prediction also requires assumptions about saving, investment, income, and prices.

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Appendix: Eaton–Kortum trade shares

Countries differ in technology T_i, production cost c_i, and iceberg trade costs $d_{ni} \geq 1$. The productivity-dispersion parameter is $\theta > 0$.

$$\pi_{ni} = \frac{T_i (c_i d_{ni})^{-\theta}}{\sum_j T_j (c_j d_{nj})^{-\theta}}, \qquad X_{ni} = \pi_{ni} X_n.$$

- Buyers source each good from the lowest delivered-cost producer. Aggregating across goods yields a gravity-like expenditure share.
- In the simple one-factor version, $c_i = w_i$ and labor-market clearing requires $w_i L_i = \sum_n X_{ni}$. Under balanced trade, $X_n = w_n L_n$.
- With exogenous net transfers D_n, spending becomes $X_n = w_n L_n + D_n$ and $\sum_n D_n = 0$. Determining imbalances endogenously requires saving and investment decisions.

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Eaton & Kortum (2002): Ricardian Microfoundations

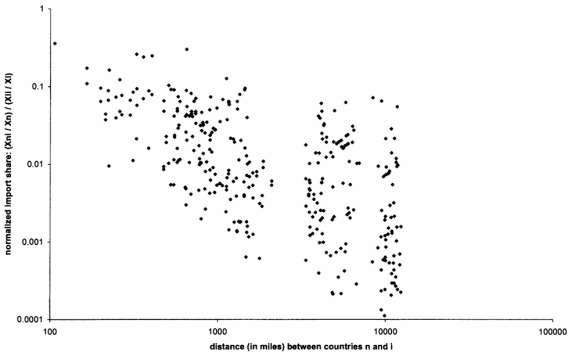


FIGURE 1.—Trade and geography.

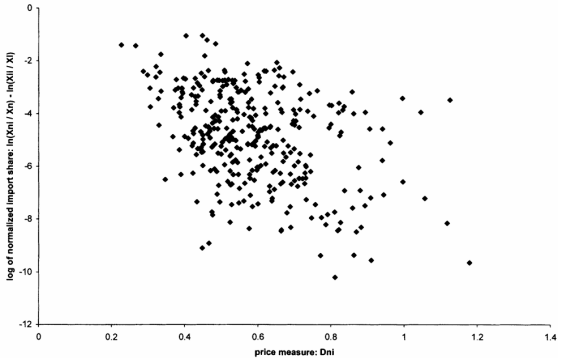


FIGURE 2.—Trade and prices.

Historical exhibit. Data period, coverage, and units are as labeled.

Source: Eaton & Kortum (2002).

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Appendix: The Home Expenditure Share and Welfare

In the one-factor model with no intermediate goods, let

$$\Phi_n = \sum_i T_i (w_i d_{ni})^{-\theta}, \quad \pi_{nn} = \frac{T_n w_n^{-\theta}}{\Phi_n}, \quad P_n = \gamma \Phi_n^{-1/\theta},$$

where γ is a positive constant determined by the distribution and preferences, and $d_{nn} = 1$.

$$\frac{w_n}{P_n} = \gamma^{-1} T_n^{1/\theta} \pi_{nn}^{-1/\theta}.$$

- Holding technology and model parameters fixed, a lower home expenditure share raises real income in this benchmark.
- Higher productivity also raises real income.
- Balanced-trade market clearing determines relative wages up to a numeraire (Walras' law). International transfers change spending and the equilibrium.

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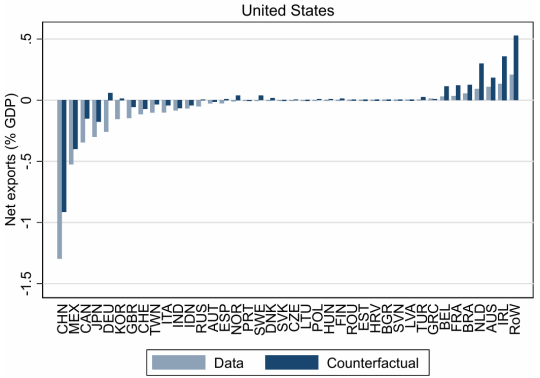
Appendix: Gravity, Welfare, and Bilateral Imbalances

- In a simple one-sector benchmark, holding technology fixed, proportional real-income changes satisfy $\widehat{W} = \widehat{\lambda}_{nn}^{-1/\theta}$, where λ_{nn} is the domestic expenditure share.
- A lower domestic share can signal gains from access to foreign supply within that model. The formula is not a model-free welfare rule and does not determine the current account.
- Cuñat and Zymek study bilateral imbalances in a dynamic trade framework. Aggregate net borrowing, triangular production chains, and asymmetric trade frictions are different sources of bilateral deficits.
- Financial autarky can restrict aggregate imbalances while bilateral deficits persist through triangular trade. Numerical decompositions require the paper's exact sample, normalization, and counterfactual.

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Cuñat & Zymek (2024)

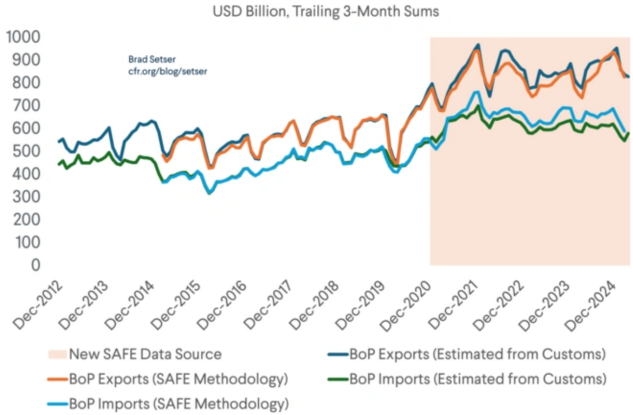
Figure 9: U.S. bilateral net exports under financial autarky



Financial-autarky counterfactual: bilateral deficits persist even when aggregate balances are zero. The reported fitted slope is 0.84; this does not mean every deficit falls by 16%. Source: Cuñat and Zymek (2024), Section 3.4.

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China's Balance of Payments: Trade Perspective

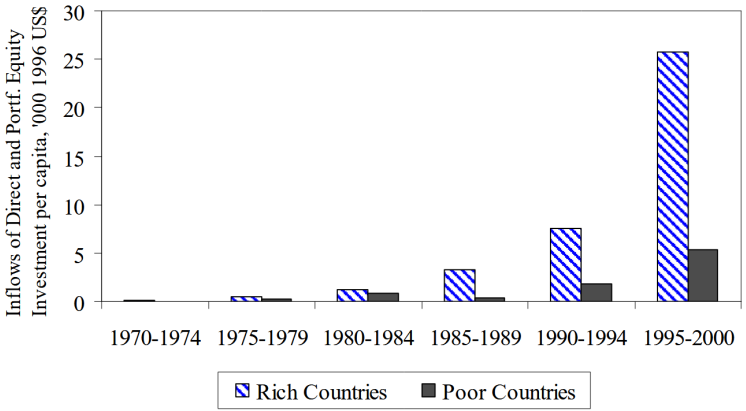


Historical exhibit. Data period, coverage, and units are as labeled.

Source: Setser.

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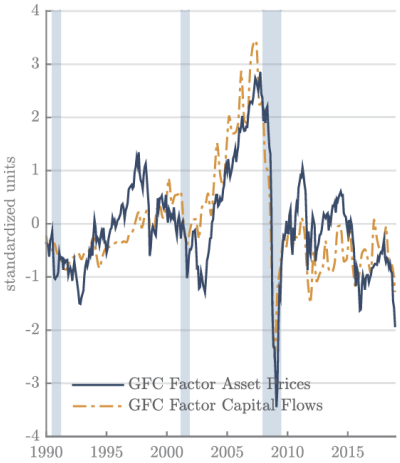
Lucas Paradox: Does Capital Flow from Rich to Poor Countries?



Historical exhibit. Data period, coverage, and units are as labeled.

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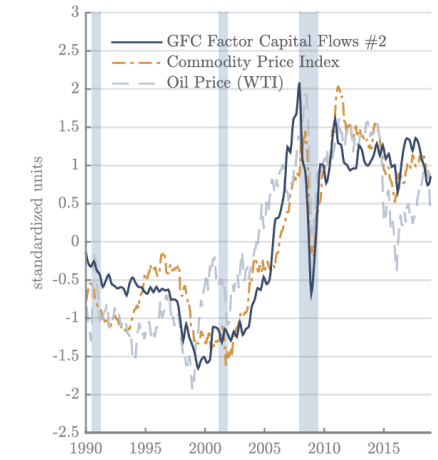
Global Financial Conditions, Asset Prices, and Volatility

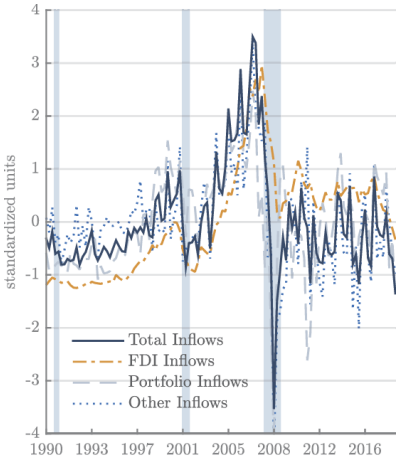


Historical exhibit. Data period, coverage, and units are as labeled.

Source: Miranda-Agrippino & Rey (2022).

Trade and Financial Cycles in Capital Flows

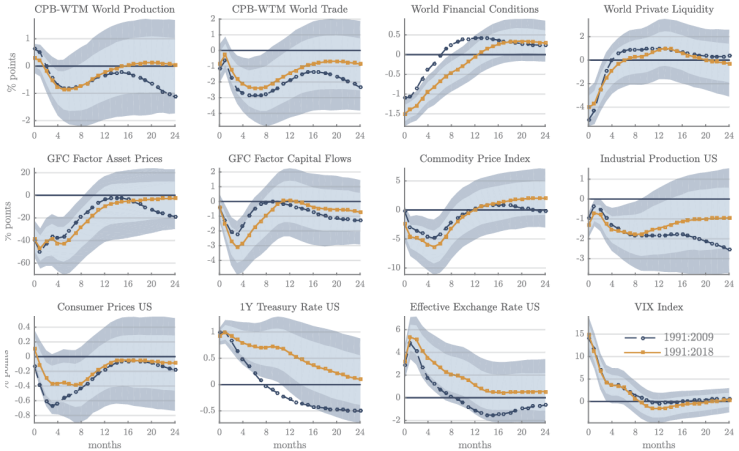




Source: Miranda-Agrippino & Rey (2022).

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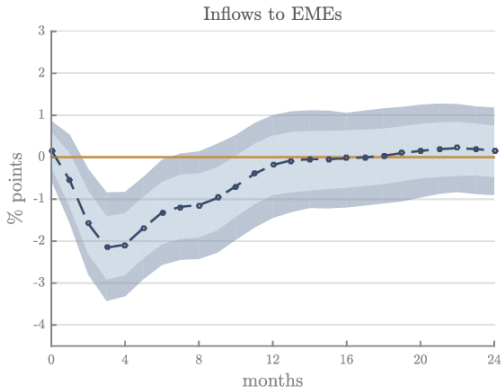
Global Financial Conditions after US Monetary Tightening

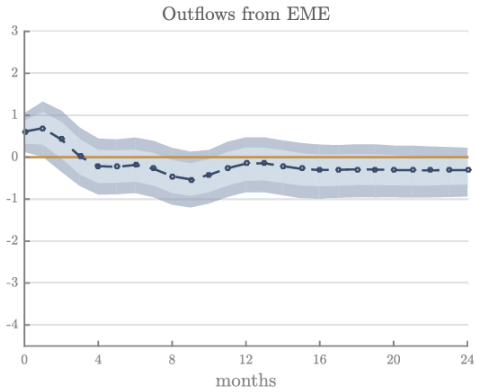


Source: Miranda-Agrippino & Rey (2022).

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Financial Flows to and from Emerging Markets





Historical exhibit. Data period, coverage, and units are as labeled.

Source: Miranda-Agrippino & Rey (2022).

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